194.155 Sustainability in Computer Science

Computer Science Between Energy Savings and Waste: From Videoconferencing to Blockchain

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Abstract

This paper examines sustainability challenges in computer science, focusing on the relationship between system architecture and energy consumption. Using blockchain technology as a central case study, it analyses how Proof-of-Work consensus mechanisms contribute to large scale electricity demand and how alternative approaches such as Proof-of-Stake can significantly reduce operational energy use. The analysis compares blockchain systems with other digital infrastructures, including data centres and videoconferencing services, and highlights structural differences in how energy demand scales.

Beyond operational electricity consumption, the paper considers lifecycle emissions, artificial intelligence scalability, hardware impacts, and governance initiatives. The findings suggest that sustainability in computer science is fundamentally shaped by architectural design choices, incentive structures, and long term system dynamics rather than by efficiency improvements alone.

Contents

1 Research Questions

1.1 Introduction

Blockchain technology has emerged as one of the most controversial innovations in contemporary computer science. Originally introduced through the publication of the Bitcoin whitepaper by Satoshi Nakamoto in 2008, titled *Bitcoin: A Peer to Peer Electronic Cash System*,blockchain proposed a decentralized system for secure, trustless digital transactions without reliance on centralized intermediaries [1].

Since then, blockchain systems have expanded far beyond cryptocurrency and are now applied in decentralized finance (DeFi), supply chain tracking, healthcare data management, digital identity systems, and governance infrastructures [2, 3]. These applications rely on distributed consensus mechanisms and cryptographic validation to ensure immutability, transparency, and resistance to tampering.

However, these security guaranties come at a substantial computational cost. Particularly Proof of Work (PoW) based systems such as Bitcoin require continuous competitive computation to validate blocks, leading to significant electricity consumption.

According to the Cambridge Bitcoin Electricity Consumption Index (CBECI), Bitcoin's annual electricity consumption rose from approximately 8–10 TWh in 2019 to a peak of about 174 TWh in 2021–2022, with modeled upper bound estimates exceeding 200 TWh. [4]

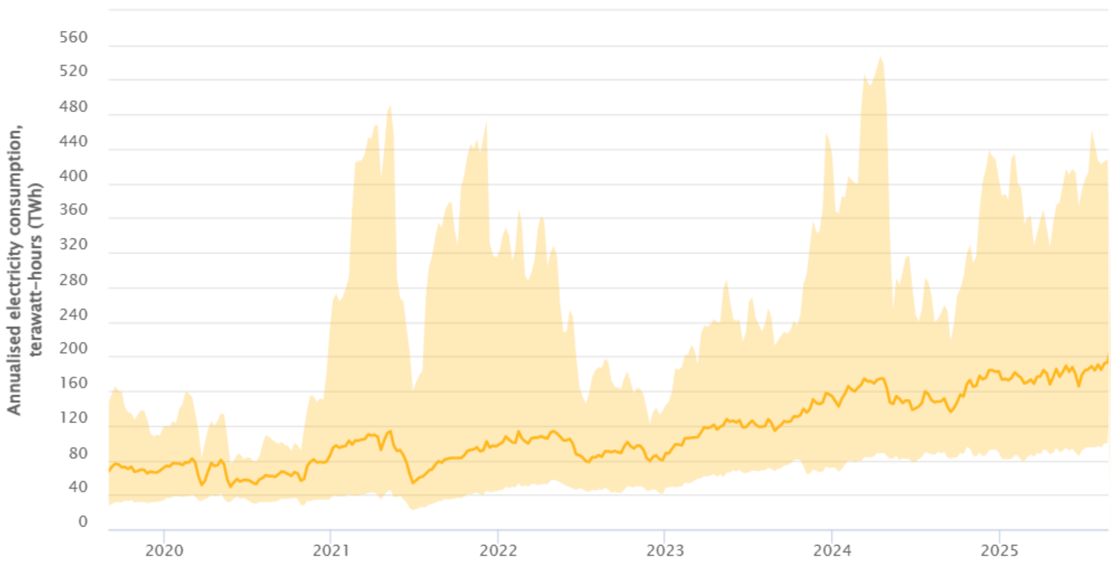

Figure 1: Annualized Bitcoin electricity consumption (TWh). Source: Cambridge Bitcoin Electricity Consumption Index (CBECI), Cambridge Centre for Alternative Finance, accessed 20 February 2026.

The rapid growth in Bitcoin's electricity consumption shown in Figure 1 indicates that the environmental impact of blockchain systems is not marginal but structurally embedded in the design of Proof-of-Work (PoW) consensus mechanisms [5–7].

Unlike other digital services, whose energy use primarily scales with user demand and efficiency improvements in data centers [8, 9], Bitcoin's PoW protocol encourages continuous competitive computation, causing energy consumption to increase with mining difficulty rather than transaction volume [5,10]. This dynamic has led to substantial carbon emissions and growing electronic waste, raising sustainability concerns [6, 11]. Given the urgency of global climate mitigation emphasized by the IPCC [12], architectural decisions in computer systems must be understood as climate relevant choices. Consequently, it becomes necessary to examine whether alternative consensus mechanisms, such as Proof-of-Stake (PoS), can maintain decentralization while significantly reducing environmental impact [13–15]. This tension between decentralization, security, and environmental sustainability leads to the central research question of this paper.

Central Research Question:

> To what extent do blockchain consensus mechanisms and cryptographic security architectures contribute to global energy consumption, and how effective are existing mitigation strategies in reducing their environmental impact?

To address this overarching question, four sub questions are examined.

1.2 How Does Blockchain Energy Consumption Compare to Other Digital Services?

The energy intensity of blockchain systems is most prominently associated with Proof of Work (PoW) consensus. In PoW based networks such as Bitcoin, miners compete to solve cryptographic hash puzzles in order to append new blocks to the chain. The protocol dynamically adjusts mining difficulty to maintain approximately constant block intervals, which implies that total network energy expenditure is largely determined by mining profitability rather than transaction volume [1]. As a result, energy demand scales primarily with economic incentives and market valuation instead of functional usage.

Building on the previously established scale of Bitcoin's electricity demand,(see p. 3) Bitcoin operates at a three digit terawatt hour scale of electricity demand [4]. Even the lower bound of current estimates represents macro scale energy consumption. For contextual comparison, the International Energy Agency (IEA) estimates that global data centers (excluding cryptocurrency mining) consumed approximately 200-240 TWh in 2022, while global data transmission networks required an additional 250 TWh [9]. Thus, a single decentralised financial network consumes electricity on the same order of magnitude as a substantial fraction of the world's entire data centre infrastructure.

At the level of individual digital services, however, energy demand appears considerably lower. IEA assessments indicate that one hour of HD videoconferencing with approximately five participants requires roughly 0.10 kWh in network transmission energy [9]. Similarly, one hour of Netflix streaming corresponds to approximately 0.077 kWh of network transfer energy [16].

These figures primarily reflect energy use associated with data transmission and exclude end user device electricity consumption, as IEA assessments distinguish between network infrastructure, data centers, and consumer hardware [9]. Device level energy demand varies depending on hardware type, screen size, and usage conditions. For example, electronic display regulations indicate that large television screens operate at substantially higher on mode power levels than smaller portable devices such as smartphones [17]. Even when such device consumption is taken into account, the per user hourly electricity demand of videoconferencing or streaming services remains negligible compared to the aggregate, system-level electricity demand associated with Proof-of-Work (PoW) consensus mechanisms, which operates at a three digit terawatt hour scale annually [4–6].

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Activity	Energy Consumption (Approx.)
1h HD group video call (5 participants)	~ 0.10 kWh
1h Netflix streaming (network transfer only)	~ 0.077 kWh
Estimated energy per Bitcoin transaction	1375–1444 kWh
Annual global data centers (excl. crypto)	200–240 TWh
Annual Bitcoin network	100–200 TWh

Table 1: Approximate comparison of energy consumption across digital activities based on estimates from the International Energy Agency [9, 16], de Vries [5], and the Cambridge Bitcoin Electricity Consumption Index [4].

The estimated range of 1375–1444 kWh per Bitcoin transaction is obtained by dividing mid range annual network electricity consumption (approximately 150–160 TWh according to CBECI [4]) by the corresponding annual number of processed Bitcoin transactions. This calculation follows the approach discussed by de Vries [5]. However, the resulting "energy per transaction" metric is methodologically controversial. Because total network energy consumption is primarily determined by mining difficulty and expected block rewards, reducing transaction throughput does not proportionally reduce total electricity demand. The metric therefore reflects an average allocation of system wide security costs rather than a marginal energy requirement for a single transaction.

A comparison of annual figures further illustrates the structural difference between blockchain systems and conventional ICT services. While global data centers support a vast range of digital applications including cloud computing, search engines, social media, scientific simulations, and enterprise systems, the Bitcoin network performs a comparatively narrow financial function. Yet its annual electricity demand represents roughly 40–100% of the electricity consumed by global data centers (excluding cryptocurrency mining), depending on the specific year and estimate used [4, 9]. This proportional relationship highlights the macroeconomic relevance of consensus design in shaping digital sustainability. The key distinction lies on scaling. Videoconferencing and streaming platforms scale energy consumption primarily with user demand, data throughput, and hardware efficiency [9]. Improvements in data centre cooling technologies, server efficiency, network optimisation, and increased integration of renewable energy have contributed to stabilising or moderating emissions growth despite rising data traffic [9,16]. Energy consumption in these systems is therefore largely proportional to service demand and technological efficiency gains.

In contrast, PoW based blockchain networks embed continuous computational competition directly into their security architecture [1]. Because mining rewards are economically determined, increases in asset valuation incentivise additional mining capacity, leading to higher aggregate hash rates and corresponding electricity consumption [4, 5]. Thus, Energy demand in PoW systems is structurally linked to market dynamics and security incentives rather than purely to functional transaction throughput.

This comparison demonstrates that consensus mechanism design is not merely a technical implementation choice but determines system level environmental impact. Blockchain sustainability must therefore be evaluated not only in terms of how much electricity is consumed but also in terms of the architectural logic that governs how and why that energy is expended [5].

1.3 What Is the Environmental Impact Beyond Electricity Consumption?

The environmental impact of electricity consumption depends on the type of energy sources used to generate it. Bitcoin mining has historically concentrated in regions with low electricity prices, often linked to coal based generation, although geographic shifts toward renewable heavy regions have also been observed [6].

Beyond operational electricity use, blockchain systems also generate hardware related environmental costs. Application Specific Integrated Circuits (ASICs), used for PoW mining, have short economic lifespans due to increasing mining difficulty. This contributes to electronic waste [11].

From a climate policy perspective, such emissions are relevant in the broader context of limiting global warming to 1.5°C above pre-industrial levels. [12]The environmental assessment of blockchain must therefore consider:

- Electricity use
- Carbon intensity of energy sources
- Hardware emissions
- Infrastructure cooling and maintenance

1.4 How Do Security Mechanisms Contribute to Energy Overhead?

Blockchain systems rely on cryptographic primitives such as hash functions, digital signatures, and public key infrastructures to ensure integrity and tamper resistance [18]. In most distributed systems, these operations are computationally lightweight relative to overall system activity. However, in Proof-of-Work (PoW) blockchains, security is achieved through continuous large-scale computation.

Bitcoin implements a PoW mechanism inspired by Adam Back's Hashcash system [1, 19]. Miners repeatedly compute SHA-256 hashes of block data combined with a variable nonce until the resulting hash satisfies a predefined difficulty target. Because cryptographic hash outputs are effectively pseudorandom, the probability of finding a valid solution is extremely low, requiring vast numbers of repeated computations.

Crucially, the protocol dynamically adjusts the mining difficulty to maintain a roughly constant block production rate despite changes in total network computing power [1, p. 3–4]. As a result, increases in hardware efficiency or participation do not improve transaction throughput; instead, they increase the total amount of computation performed across the network. In this sense, PoW embeds energy expenditure directly into its security architecture.

Mining participation responds to economic incentives, particularly block rewards and asset valuation [5]. Higher expected profitability encourages additional computational investment, raising the global hash rate and aggregate electricity consumption [4, 6]. Consequently, energy overhead in PoW systems is not merely an implementation inefficiency but an intrinsic feature of how consensus and security are achieved.

1.5 What Mitigation Strategies Exist?

Given the substantial electricity demand associated with Proof-of-Work (PoW) consensus, mitigation strategies can be analysed along five dimensions: (1) consensus redesign, (2) hardware efficiency improvements, (3) energy sourcing and carbon intensity, (4) scaling and transaction efficiency solutions, and (5) regulatory or economic intervention.

Consensus Redesign: Proof of Stake The most structurally transformative mitigation strategy is the replacement of PoW with alternative consensus mechanisms, particularly Proof-of-Stake (PoS). In PoW systems, security is achieved through continuous computational competition, and mining participation responds to economic incentives derived from block rewards and asset valuation [1, 5]. Because mining difficulty adjusts dynamically, aggregate electricity consumption reflects expected profitability rather than transaction throughput.

PoS eliminates the need for energy intensive hash competition by selecting validators according to their economic stake. Instead of expending electricity to demonstrate commitment, validators risk losing staked assets in the case of dishonest behaviour [13]. This architectural shift removes the structural coupling between network security and electricity consumption.

The transition of Ethereum from PoW to PoS in September 2022 ("The Merge") provides empirical evidence of this effect. The Crypto Carbon Ratings Institute estimates that Ethereum's electricity demand decreased by approximately 99% following the transition [14]. Independent estimates published by the Ethereum Foundation report a reduction from tens of terawatt hours annually under PoW to well below one terawatt hour under PoS [15]. These figures indicate that consensus redesign can produce orders of magnitude reductions in operational electricity demand.

However, PoS introduces governance trade offs. Because voting influence is proportional to economic stake, concerns arise regarding wealth concentration and validator centralisation [13]. Thus, while PoS mitigates electricity consumption, it alters the distribution of control and trust assumptions within the network.

Hardware Efficiency and Rebound Dynamics A second mitigation pathway involves improvements in mining hardware efficiency. Successive generations of application specific integrated circuits (ASICs) have substantially improved hash rate per watt performance over time [5, 20]. From a technical perspective, higher efficiency per unit of computation should reduce electricity consumption if total computational activity remains constant.

However, empirical evidence suggests that aggregate network electricity consumption does not necessarily decline in proportion to hardware efficiency gains. De Vries [5] argues that efficiency improvements reduce operational costs, thereby incentivising additional mining participation. Similarly, Gallersdörfer et al. [20] show that network wide energy consumption is driven by competitive equilibrium rather than purely by device efficiency. This dynamic resembles a rebound effect: reductions in energy intensity per hash can stimulate higher total hash rates, maintaining or increasing overall electricity demand.

Data from the Cambridge Bitcoin Electricity Consumption Index indicate that total annual network electricity consumption has remained in the range of 100 200 TWh despite ongoing hardware improvements [4]. Consequently, efficiency gains at the hardware level do not automatically translate into system wide reductions in electricity demand.

Energy Sourcing and Carbon Intensity Mitigation strategies may also target the carbon intensity of electricity used for mining. The climate impact of electricity consumption depends on the underlying energy mix rather than solely on total kilowatt hours. Krause and Tolaymat [10] quantify the carbon costs associated with cryptocurrency mining and demonstrate that environmental impact varies significantly with regional energy sources. Similarly, Stoll et al. estimate substantial variation in Bitcoin's carbon footprint depending on geographic distribution and electricity generation profiles [6].

Some industry actors argue that mining can utilise surplus renewable generation that would otherwise be curtailed. However, empirical assessments indicate that mining activity is often concentrated in regions with low electricity prices, which may include both renewable rich areas and fossil dependent grids [5, 6]. While renewable integration reduces emissions per kilowatt hour, it does not eliminate the underlying structural demand for electricity inherent in PoW systems.

Layer 2 Scaling and Transaction Efficiency Layer 2 scaling solutions, such as the Lightning Network, aim to increase transaction throughput by processing smaller transactions off chain and periodically settling balances on the base layer. These mechanisms improve functional efficiency by increasing the number of transactions supported per unit of on chain activity.

However, because base layer security in PoW systems remains dependent on computational mining, total network electricity consumption continues to be governed primarily by mining incentives rather than transaction count [5]. Thus, Layer 2 scaling enhances transaction efficiency but does not fundamentally alter the energy architecture of PoW consensus.

Regulatory and Economic Instruments Finally, mitigation may occur through regulatory or market based instruments. Carbon pricing, environmental disclosure requirements, or electricity taxation could internalise environmental externalities and thereby alter mining profitability. Since PoW energy demand responds to economic incentives embedded in the protocol design [1, 5], policy interventions that affect electricity costs could indirectly influence total hash rate and associated electricity consumption.

Taken together, these mitigation strategies differ significantly in their structural impact. The transition from PoW to PoS offers the most substantial reduction in absolute electricity demand [14, 15]. In contrast, improvements in mining hardware efficiency primarily reduce energy intensity per unit of computation, but empirical evidence suggests that aggregate electricity consumption remains driven by competitive equilibrium and mining profitability [4, 5, 20]. Similarly, shifting toward renewable energy sources reduces carbon intensity per kilowatt hour but does not eliminate the underlying structural dependence of PoW security on continuous computation [6, 10]. Consequently, existing literature indicates that the long-term sustainability of blockchain systems depends fundamentally on consensus architecture rather than incremental efficiency or sourcing improvements alone [1, 5].

1.6 Synthesis

The analysis of blockchain energy consumption demonstrates that sustainability outcomes are fundamentally shaped by consensus architecture rather than by isolated efficiency metrics. Proof of Work systems embed continuous computational competition into their security model, linking electricity demand directly to mining profitability and asset valuation [1, 5]. As empirical data from the Cambridge Bitcoin Electricity Consumption Index indicates, this design results in annual electricity consumption on the order of 100–200 TWh [4], comparable to substantial segments of global data centre infrastructure [9].

In contrast, conventional digital services such as videoconferencing and streaming scale primarily with user demand and technological efficiency improvements [9, 16]. While these services also contribute to global electricity consumption, their energy demand is functionally tied to usage intensity and hardware efficiency rather than to competitive security incentives. This structural difference explains why incremental improvements in codecs, cooling systems, and server efficiency can moderate emissions growth in ICT infrastructure, whereas similar efficiency gains in PoW systems do not necessarily reduce total electricity demand.

Mitigation strategies further reinforce this distinction. Hardware efficiency improvements and renewable energy sourcing primarily affect energy intensity or carbon intensity [6, 10]. However, they do not eliminate the underlying requirement for continuous hash based competition. By contrast, consensus redesign most prominently the transition from PoW to PoS demonstrates that secure distributed validation can be achieved without large scale electricity expenditure, as evidenced by Ethereum's post Merge energy reduction [13–15].

Taken together, the findings suggest that blockchain sustainability is not merely a question of operational optimisation but of architectural choice. The environmental impact of decentralised systems emerges from the incentive structures embedded in their protocol design. Consequently, evaluating blockchain technologies from a sustainability perspective requires analysis at the level of consensus mechanisms, economic incentives, and security assumptions rather than solely at the level of hardware efficiency or carbon sourcing.

The central research question regarding the extent to which blockchain consensus mechanisms contribute to global energy consumption and the effectiveness of mitigation strategies can therefore be answered as follows:

> Proof-of-Work architectures contribute substantially to global electricity demand due to their structurally competitive validation model. While mitigation strategies can reduce carbon intensity and improve efficiency, only consensus redesign fundamentally alters the energy profile of blockchain systems. Sustainability in distributed computing thus depends primarily on how security is achieved and how incentives are structured within the protocol itself.

2 Sustainability in Computer Science: A Broader Perspective

The presentation by Mayrhofer [21] highlighted several central sustainability issues in computer science, including the energy consumption of videoconferencing services, data centre infrastructure, blockchain systems, and the overhead introduced by security mechanisms. These examples illustrate the tension between digital efficiency gains and resource intensive system design.

Building on the themes introduced in the presentation, this section broadens the perspective to examine additional dimensions such as scalability in Artificial Intelligence(AI), hardware life cycle emissions, green software engineering practices, governance initiatives, and systemic rebound effects [9, 22, 23].

By situating blockchain and digital communication within a wider structural and life cycle context, the analysis aims to demonstrate that sustainability in computer science is not limited to individual technologies, but also infrastructure design, economic incentives, material production, and long term policy alignment.

2.1 Data centers and Global ICT Infrastructure

Beyond blockchain systems, global data centre infrastructure represents a central pillar of digital sustainability. Data centres support cloud computing, search engines, streaming platforms, enterprise systems, and increasingly artificial intelligence workloads. According to the International Energy Agency, global data centres consumed approximately 200–240 TWh of electricity in 2022, while data transmission networks required an additional ∼250 TWh [9]. Together, these systems account for roughly 2–3% of global electricity demand [9, 22].

Despite rapid growth in digital services, several studies demonstrate that efficiency improvements have moderated electricity growth in this sector. Masanet et al. show that between 2010 and 2018, global data centre workloads increased more than sixfold, while electricity consumption rose only modestly, largely due to server virtualisation, hyperscale optimisation, and improved cooling technologies [8]. Similar findings are reported by Freitag et al., who highlight significant efficiency gains across network infrastructure and hardware performance [23]. These developments illustrate that architectural optimisation and infrastructure consolidation can partially decouple computational growth from proportional energy increases.

However, long term sustainability remains uncertain. Andrae and Edler project substantial growth in ICT electricity consumption under high demand scenarios, particularly if efficiency gains plateau [24]. More recent assessments by the IEA suggest that artificial intelligence and high-performance computing may significantly increase electricity demand due to energy-intensive GPU clusters and large-scale model training [9]. Studies on machine learning energy consumption further indicate that training large neural networks can generate substantial carbon emissions depending on energy sourcing [25, 26].

These dynamics highlight a structural tension within digital infrastructure. While engineering innovation has historically improved energy efficiency, rising computational demand, expanding cloud services, and AI scaling introduce rebound risks [23]. Consequently, the sustainability of global ICT infrastructure depends not only on technological efficiency improvements but also on demand governance, energy sourcing, and system-level optimisation.

2.2 Artificial Intelligence and Computational Scaling

Artificial intelligence represents another sustainability dimension not deeply addressed in the presentation. Training large machine learning models requires extensive computational resources and electricity intensive data centre infrastructure [25]. Strubell et al. estimate that training certain deep learning models can generate substantial carbon emissions depending on energy sourcing [25]. Henderson et al. emphasise the need for systematic reporting of energy and carbon footprints in machine learning research [26].

Unlike Proof of Work systems, AI does not structurally require competitive energy expenditure for security [25]. However, the scale of training runs, hyperparameter optimisation, and inference deployment can lead to large cumulative energy demand [9]. Sustainability challenges in AI therefore concern growth and scale rather than incentive driven waste.

2.3 Hardware Production, Lifecycle Emissions and Electronic Waste

Another major sustainability topic in computer science concerns embodied emissions and hardware life cycle impacts. Belkhir and Elmeligi estimate that a substantial portion of ICT related emissions arises from device manufacturing rather than operational electricity use [22]. Semiconductor fabrication, material extraction, and device assembly are energy intensive processes contributing to global emissions [22].

Electronic waste represents a related challenge. De Vries and Stoll highlight the short economic lifespan of ASIC mining hardware, which contributes to growing electronic waste streams [11]. More broadly, rapid device turnover cycles in consumer electronics increase environmental pressure [23]. Extending hardware lifespans and promoting circular economy principles therefore represent critical sustainability strategies [23].

2.4 Green Software Engineering and Energy Aware Design

Sustainability is not limited to hardware and infrastructure but also concerns software design. Energy efficient algorithms, reduced data transfer, and optimised memory usage can significantly reduce operational electricity demand [23]. The Green Software Foundation promotes principles such as carbon aware computing and energy proportional software development [27]. These initiatives demonstrate meaningful progress in integrating sustainability principles into software engineering practice.

However, efficiency improvements may not automatically reduce total emissions due to rebound effects [23]. Increased efficiency can lower costs and expand digital service usage, potentially offsetting environmental gains [5, 23]. This highlights the importance of considering systemic demand alongside technical optimisation.

2.5 Digitalisation as a Sustainability Enabler

Computer science also enables sustainability improvements in other sectors. The International Energy Agency emphasises the role of digital technologies in optimising energy systems, improving smart grid efficiency, and enhancing industrial monitoring [9]. Videoconferencing and remote collaboration tools can reduce transportation related emissions under certain usage conditions [9].

Freitag et al. argue that ICT can have transformative climate impact if deployed strategically and governed responsibly [23]. Thus, there could be a possibility in which computer science simultaneously contributes to environmental pressure and offers tools for mitigation.

2.6 Relative Impact Compared to Other Industries

Compared to heavy industry and transportation, ICT accounts for a smaller share of global emissions, estimated at approximately 2–3% [22, 23]. Transportation and industrial sectors contribute substantially larger shares of global greenhouse gas emissions [12]. Nevertheless, ICT's rapid growth rate and its enabling role in economic systems amplify its longterm sustainability relevance [23].

Moreover, as demonstrated by energyintensive blockchain systems, architectural choices within computer science can generate disproportionate environmental impact relative to functional output [4, 5]. Therefore, sustainability in computer science is not solely about sectoral size but about design responsibility.

2.7 Organisations and Governance Initiatives

Several organisations actively promote sustainability in computer science. The International Energy Agency provides ongoing assessments of digital infrastructure energy demand [9]. The Green Software Foundation develops standards and best practices for reducing softwarerelated emissions [27]. Academic research initiatives increasingly require disclosure of computational energy use in machine learning publications [26].

These developments indicate meaningful institutional progress toward integrating sustainability metrics into digital system design and governance.

Overall, Sustainability in computer science encompasses infrastructure efficiency, AI scalability, software optimisation, hardware lifecycle management, and governance mechanisms [9, 22, 23]. Significant achievements include improved data centre efficiency [8], renewable energy procurement [9], and consensus redesign in blockchain systems [14]. However, continued growth in computational demand and rebound dynamics present ongoing challenges [23, 24].

Ultimately, the environmental impact of computer science depends not only on technological efficiency but also on architectural design choices, policy frameworks, and responsible deployment strategies aligned with global climate targets [12, 23].

3 Reflection on Sustainability and Personal Perspective

Working on this seminar paper significantly changed how I understand sustainability in computer science. At the beginning of the course, I primarily associated sustainability with visible issues such as renewable energy use or the electricity consumption of data centres. The presentation by Mayrhofer [21] presented concrete examples including the energy demand of videoconferencing, global data centre infrastructure, blockchain systems, and security related computational overhead. Seeing these examples side by side made it clear that different digital architectures can have very different environmental implications.

The subsequent literature research helped me understand these differences more precisely. Studies on Proof-of-Work blockchains show that high electricity consumption is structurally linked to the consensus mechanism and its economic incentives [5, 6]. This means that energy use is not merely the result of inefficient implementation but an inherent feature of how security is achieved. In contrast, research on data centres demonstrates that efficiency improvements in hardware, cooling, and workload management have historically moderated electricity growth despite increasing demand [8]. Comparing these two cases strengthened my understanding that sustainability outcomes depend strongly on architectural design choices.

Another important insight concerns lifecycle impacts. Before writing this paper, I focused mainly on operational electricity use. Literature on embodied emissions and electronic waste broadened my perspective by showing that environmental impacts also arise during hardware production and disposal [22, 23]. This shifted my attention towards long term system design, material use, and device longevity.

ChatGPT served as a research support tool during the research process. However, it quickly became clear that suggested formulations and numerical claims required independent verification through peer reviewed articles and official reports. Cross checking sources and validating references improved my critical reading skills and reinforced the importance of academic research. The final responsibility for accuracy remained mine.

Overall, this seminar paper changed my understanding of sustainability in computer science from a narrow focus on electricity consumption toward a broader perspective that includes consensus design, incentive structures, lifecycle emissions, infrastructure scaling, and governance. I now view sustainability not as an afterthought, but as a design constraint that must be considered from the outset of system development.

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